

Total No. of Questions : 10]

SEAT No. :

**P2621**

**[5153]-597**

[Total No. of Pages : 4

**T.E. (I.T.)**

**SYSTEMS PROGRAMMING**

**(2012 Course) (Semester - II) (314450)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Give the various data structures in the design of pass-1 of a Two-pass direct linking loader. **[4]**
- b) What are the assembler directives? Explain how assembler directives LTORG, ORIGIN and EQU are processed in first pass. **[6]**

OR

- Q2)** a) For the 'C' code given below, give the different tables that would be generated as output of lexical analysis. **[8]**

```
main ()
{
    Float average ;
    int i, sum , n = 10;
    sum = 0;
    clrscr();
    printf("Average of 10 nos. : %f",avg);
    for (i = 1; i <= 10; i++)
        Sum = sum +i;
    avg = sum / (float) n;
}
```

- b) Define macroprocessor and assembler. **[2]**

**P.T.O.**

- Q3) a)** For the following piece of assembly language code, show the contents of MDT, MNT, IC and EC, [5]

```
MACRO
INCR & ARG
LOAD 1,&ARG
ADD 1, = F'1'
STORE 1,&ARG
MEND
MACRO
DECR & NUM
LOAD 2, &NUM
SUB 2, = F'1'
STORE 2, & NUM
MEND
START
DECR D1
STORE AREG, D1
INCR D2
D1 DC '50'
D2 DC '100'
END
```

- b) Define loader and enlist the basic functions of loader. [5]

OR

- Q4) a)** Using the algorithm convert the following regular expressions to DFA:  
(a.b)\*.a.# [6]
- b) Explain different parameter passing mechanisms in macro-processor. [4]

**Q5) a)** Consider the grammar [6]

$$E \rightarrow E-E$$

$$E \rightarrow E/E$$

$$E \rightarrow id$$

Perform shift Reduce parsing of i/p string “id – id/id”

b) Explain recursive descent parser for the given grammar to derive the string cad

$$S \rightarrow cAd$$

$$A \rightarrow ab/a$$

[6]

c) Compare bottom UP and top down parser.

[6]

OR

**Q6) a)** Consider the following grammar [10]

$$S \rightarrow S(S)S/\epsilon$$

Construct SLR parser and parse for the string (a,(a,a))

b) Explain YACC file structure. [4]

c) Compare SLR, CLR and LALR parsers. [4]

**Q7) a)** Write down Syntax Directed Translation for assignment statement. [6]

b) For the grammar [6]

$$D \rightarrow TL$$

$$T \rightarrow \text{int/real}$$

$$L \rightarrow L, \text{id/id}$$

Draw an annotated parse tree for the statement  $\text{real } x_1, x_2;$

c) Write the method of generating intermediate code for the expression

If (condition) then  $p = q$  Else  $x = y + z$  [4]

OR

- Q8)** a) Define Syntax directed definition and syntax directed translation. [4]  
b) Design dependency graph for the following grammar [6]

$$E \rightarrow E+T/T$$

$$T \rightarrow T*F/F$$

$$F \rightarrow id$$

The expression given is :  $5*8-10$

- c) For the following expression write its postfix expression, draw DAG and write three address code: [6]

$$((x+y)-((x+y)*(x-y)))+(x+y)*(x-y))$$

- Q9)** a) Compare between static, stack & heap allocation. [4]  
b) With examples explain code generation issues. [6]  
c) What are the different techniques of storage allocation. [6]

OR

- Q10)** a) With examples explain at least four machine independent code optimization techniques. [8]  
b) Which are the machine dependent code optimization issues. [8]

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